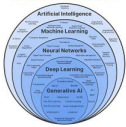


The World of
Artificial Intelligence



Probability

Basics of Probability & Statistics



EXPERIMENT

Deterministic Experiment



(Genetic determination)

Examples:

- Cooling water below 0 degrees Celsius: It will always freeze.
- Calculating the product of 2 multiplied by 3: The result is always 6.
- Throwing a ball in the sky: It will always fall down due to gravity.

Random Experiment



(Hitting the target)

• Examples:

- **Flipping a coin:** The outcome can be either heads or tails, and it's impossible to predict which will occur.
- **Rolling a dice:** The outcome can be any number from 1 to 6, and it's impossible to predict the exact number.
- **Drawing a card from a deck:** The outcome can be any of the 52 cards, and you cannot be certain which card will be drawn.



Prediction

Experiment/Trial=Possibility
of a cricket match or playing
a cricket match

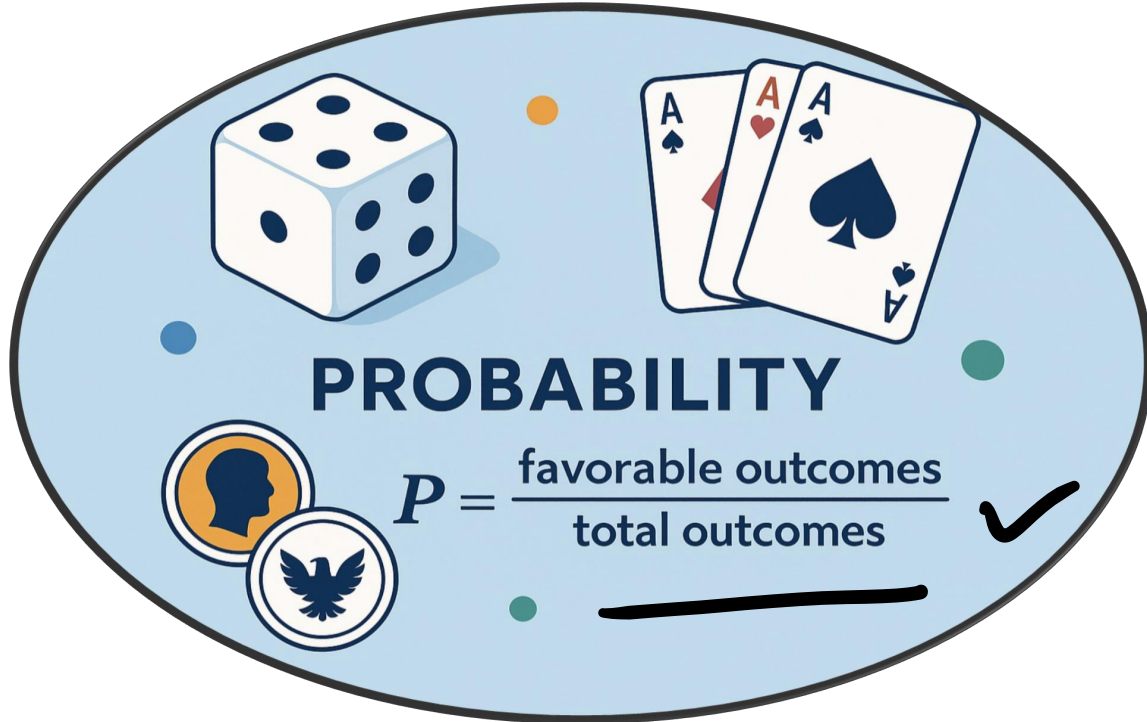


Probability



Outcome = Win, Loss

Sample Space $S = \{\text{Win, Loss}\}$



Sample Space: The sample space is the set of all possible outcomes of a random Experiment. It is denoted by S . Elements of sample space referred as Sample points.



Sample space for tossing a fair or unbiased coin (सिक्का) :

i) Tossing a single coin or unbiased coin

$S = \{H, T\}$
No. of Sample points
= 2



$= \frac{1}{2}$
 $= \frac{1}{2}$
2



Biased or unfair Coin Sholay Movie(1975)

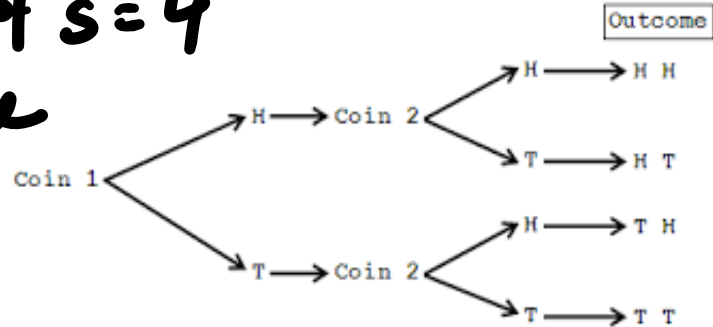


ii) When Two Coins are Tossed:

$$S = \{HH, HT, TH, TT\}$$

No. of $S = 4$

Tree



Example of
Biased
coin

→ 00-01-10-11
Binary style

iii) When Three Coins
are Tossed:

$$S = \{ HHH, HHT, HTH, HTT, THH, THT, TTH, TTT \}$$

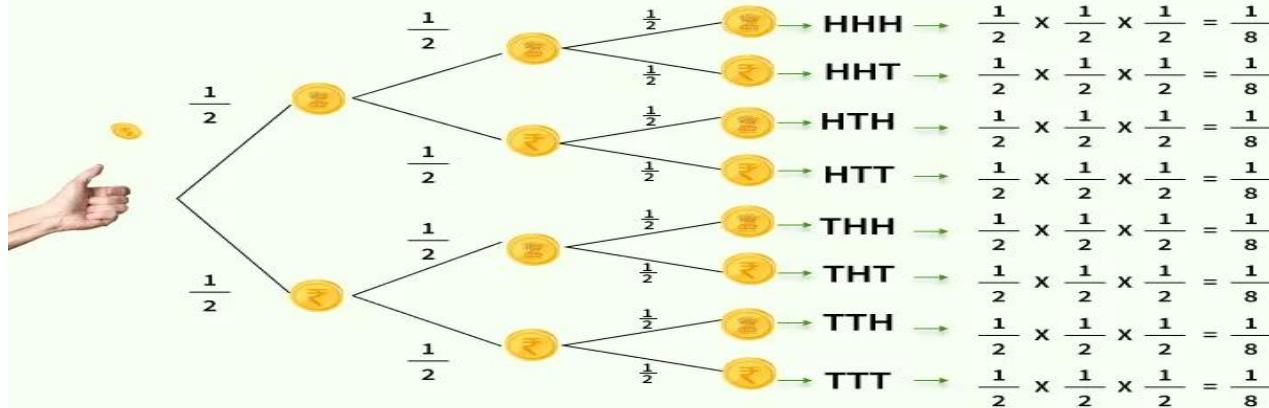
No. of sample points = 8

0	0	0
0	0	1
0	1	0
0	1	1

1	0	0
1	0	1
1	1	0
1	1	1



Coin Toss Probability



To Determine The Probability Of A Specific Outcome In Multiple Coin Tosses, Multiply The Individual Probabilities.

Coin Toss Probability Formula = (Number of Favourable Outcomes) / (Total Possible Outcomes)

iv) In general for tossing of n -coins $\rightarrow S = 2^n$

Where n = Number of coins or number of tossing

$$n = 1 \text{ coin, } S = 2^1 = 2$$

$$n = 2 \text{ coin, } S = 2^2 = 4$$

$$n = 3 \text{ coin, } S = 2^3 = 8$$

$$n = 4 \text{ coin, } S = 2^4 = 16$$

$$n = 5 \text{ coin, } S = 2^5 = 32$$

Tossing of 2 coins
or
tossing of 1 coin two times

Example:-

Probability of impossible event:

$$P(2 \text{ Sundays in a week}) = 0$$

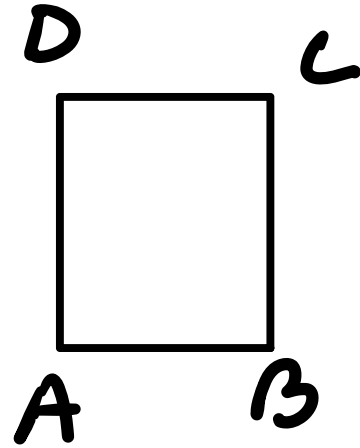
$$P(E) = 0 \quad \checkmark$$

Example:-

Probability of Sure event:

$P(4 \text{ slides in a square}) = 1$

$$P(E) = 1$$



$$0 \leq P \leq 1$$

Not possible certain

Ex ::

Two fair coins are tossed Simul.
Find the following probabilities:-

- (i) Proba. of getting two heads = 1
- (ii) Prob. of getting one head = 2
- (iii) Prob. of getting zero head. = 1

$$S = 2^n = 2^2 = 4.$$

$$S = \{HH, HT, TH, TT\}$$

$$P = \frac{\text{no. of fav. outcomes}}{\text{no. of S. points} = 4}$$

Ex :: Probability of tossing two fair coins:-

Total no of outcomes = 4

$$S = \{(H,H), (H,T), (T,H), (T,T)\}$$

Q.1 Probability of getting two heads or zero tails:

$$P(2H) = P(0T)$$

= No. of outcomes with two heads / Total outcomes

$$= \left(\frac{1}{4} \right)$$

Q.2 Probability of one head or one tail

$$P(1H) = P(1T) = \frac{\text{No. of outcomes with only one head}}{\text{Total}} \\ = \frac{2}{4} = \frac{1}{2}$$

Q.3 Probability of zero head or two tails:

$$P(0H) = P(2T) = \frac{\text{No. of outcome with two tails}}{\text{Total outcomes}} \\ = \frac{1}{4} \text{ Ans.}$$

Ex:-

Probability of tossing three fair coins:

$$\text{Total outcomes} = 2^3 = 8$$

$$S = \{(H,H,H), (H,H,T), (H,T,H), (T,H,H), (T,T,H), (T,H,T), (H,T,T), (T,T,T)\}$$

Q.1

Probability of getting zero head:

$$P(0H) = P(3T) = \frac{\text{no. of outcomes with no heads}}{\text{Total}} = \frac{1}{8}$$

Q.2

P. of getting one head or prob. of getting two tail:

$$P(1H) = P(2T) = \frac{\text{no. of outcomes with one head}}{\text{Total}} = \frac{3}{8}$$

Q.3

Prob. of getting three heads or zero tail:

$$P(3H) = P(0T) = \frac{\text{No. of outcomes with three heads}}{\text{Total}} = \frac{1}{8} \text{ Ans}$$

Example :-

A coin is tossed 4 times. What is the Probability of getting heads exactly 3 times?

$$\frac{1}{4}, \frac{3}{8}, \frac{1}{2}, \frac{3}{4}$$

M-I $S = 2^4 = 16$

$$P(x=3) = \frac{{}^4C_3}{2^4} = \frac{4}{16} = \frac{1}{4}$$

M-II

$$HHHT = \frac{4!}{3!1!} = \frac{4 \times 3!}{3! \times 1!} = 4$$

$$P(H=3) = \frac{4}{16} = \frac{1}{4}$$

Note:-

Otherwise You can Create Tree Diagram.

Example :-

If three coins are tossed Simultaneously
the probability of getting at least
One head is -

- (A) $\frac{1}{8}$ (B) $\frac{3}{8}$ (C) $\frac{1}{2}$ (D) $\frac{7}{8}$

M-I $S = 2^3 = 8$ M-II

HHH ✓

HHT ✓

HTH ✓

HTT ✓

T HH ✓

T HT ✓

TT H ✓

TTT X

$P = \frac{7}{8}$

$$\begin{aligned} P(H \geq 1) &= P(H=1) + \\ &\quad P(H=2) + P(H=3) \\ &= \frac{3}{8} + \frac{3}{8} + \frac{1}{8} \\ &= \frac{7}{8} \end{aligned}$$

M-III

Imp $P(H \geq 1) = 1 - P(H=0)$
 $= 1 - \frac{1}{8} = \frac{7}{8}$

Example

An unbiased coin is tossed three times. The probability that the head turns up in exactly two cases is -

- (A) $\frac{1}{9}$ (B) $\frac{1}{8}$ (C) $\frac{2}{3}$ (D) $\frac{3}{8}$

$$\underline{M-I} \quad S = 2^3 = 8$$

000 x
001 ✓
010 ✓
011 x
100 x
101 x
110 x
111 x

$$P(H=2) = \frac{3}{8}$$

$$\underline{M-II}$$

$$P = \frac{{}^3C_2}{2^3} = \frac{3}{8} \checkmark$$

$$\underline{M-III}$$

$$HHT \rightarrow \frac{3!}{2!1!}$$

$$= \frac{3 \times 2!}{2! \times 1!} = \textcircled{3}$$

$$P(H=2) = \frac{3}{8}$$

Example

A fair coin is tossed three times in succession. If the first toss produces a head, what is the probability that both outcomes are heads?

- (A) $\frac{1}{3}$ (B) $\frac{1}{4}$ (C) $\frac{1}{2}$ (D) $\frac{2}{3}$

$$S = 2^3 = 8$$

Modified Sample Space

HHH
HHT
HTH
HTT
THH
THT
TTH
TTT

$S^* =$ HHHX
HHT✓
HTH✓
HTTX

$$P = \frac{2}{S^*} = \frac{2}{4}$$

$$P = \frac{1}{2}$$

Problems on Coin Toss Probability

1. A coin is tossed twice at random. What is the probability of getting

(i) at least one head

(ii) the same face?



Problems on Coin Toss Probability

2. If three fair coins are tossed randomly 175 times and it is found that **three heads** appeared 21 times, **two heads** appeared 56 times, **one head** appeared 63 times and zero head appeared 35 times.

What is the probability of getting

(i) three heads, (ii) two heads, (iii) one head, (iv) 0 head.

$$\begin{array}{cccc}
 \downarrow & \downarrow & \downarrow & \downarrow \\
 P = \frac{21}{175} & P = \frac{56}{175} & P = \frac{63}{175} & P = \frac{35}{175}
 \end{array}$$



Probability



Applications of Probability in Data Science

1

Statistical inference

2

Machine learning

3

Bayesian analysis

4

Risk assessment

5

Quality control

6

Anomaly detection

Probability



Example :-

If a fair coin is tossed four times.
What is the probability that two heads
and two tails will result?

- (A) $\frac{3}{8}$ (B) $\frac{1}{2}$ (C) $\frac{5}{8}$ (D) $\frac{3}{4}$.

$$S = 2^4 = 16$$

$$P = \frac{nCr}{2^4} = \frac{4C_2}{2^4}$$

$$P = \frac{6}{16} = \left(\frac{3}{8}\right)$$

Ans

HHTT

↓ No. of favorable outcomes

$$= \frac{4!}{2!2!}$$

$$= \frac{4 \times 3 \times 2!}{2! \times 2 \times 1}$$

$$= \left(6\right) \text{ Ans}$$